Unit: 6 Unit 2: Exponents Radicals
Topic: Rationalizing Denominator

Name: ____________________
Date: ____________

Foundation (Jalapeno)

Q1. To rationalize the denominator of $\frac{1}{\sqrt{3}}$, we multiply by:

(A) $\frac{\sqrt{3}}{\sqrt{3}}$
(B) $\frac{2}{2}$
(C) $\frac{\sqrt{2}}{\sqrt{2}}$
(D) $\frac{1}{\sqrt{2}}$
(E) $\frac{\sqrt{6}}{\sqrt{6}}$

Q2. What is the rationalized form of $\frac{2}{\sqrt{5}}$?

(A) $\frac{2\sqrt{5}}{5}$
(B) $\frac{2}{5}$
(C) $\frac{\sqrt{5}}{2}$
(D) $\frac{2\sqrt{2}}{5}$
(E) $\frac{5}{2\sqrt{5}}$

Q3. A spacecraft is traveling at $\frac{2}{\sqrt{3}}$ times the speed of light. What is the rationalized form of this speed?

(A) $\frac{2\sqrt{3}}{3}$
(B) $\frac{2}{3}$
(C) $\frac{\sqrt{3}}{2}$
(D) $\frac{2\sqrt{2}}{3}$
(E) $\frac{3}{2\sqrt{3}}$

Q4. The mass of a planet is $\frac{1}{\sqrt{2}}$ times the mass of Earth. What is the rationalized form of this mass?

(A) $\frac{\sqrt{2}}{2}$
(B) $\frac{1}{2}$
(C) $\frac{2}{\sqrt{2}}$
(D) $\frac{\sqrt{2}}{\sqrt{2}}$
(E) $\frac{1}{\sqrt{2}}$

Q5. The time it takes for a spaceship to travel to Mars is $\frac{3}{\sqrt{6}}$ years. What is the rationalized form of this time?

(A) $\frac{\sqrt{6}}{2}$
(B) $\frac{3\sqrt{6}}{6}$
(C) $\frac{\sqrt{3}}{2}$
(D) $\frac{3}{2\sqrt{2}}$
(E) $\frac{\sqrt{2}}{3}$

Q6. The distance between two planets is $\frac{2}{\sqrt{8}}$ light-years. What is the rationalized form of this distance?

(A) $\frac{\sqrt{2}}{2}$
(B) $\frac{2\sqrt{2}}{4}$
(C) $\frac{\sqrt{8}}{2}$
(D) $\frac{2}{2\sqrt{2}}$
(E) $\frac{2\sqrt{8}}{8}$

Q7. A lab experiment involves a chemical reaction with a rate of $\frac{1}{\sqrt{5}}$ moles per second. What is the rationalized form of this rate?

(A) $\frac{\sqrt{5}}{5}$
(B) $\frac{1}{5}$
(C) $\frac{\sqrt{2}}{5}$
(D) $\frac{1}{\sqrt{2}}$
(E) $\frac{5}{\sqrt{5}}$

Q8. The energy released by a star is $\frac{2}{\sqrt{3}}$ times the energy released by the Sun. What is the rationalized form of this energy?

(A) $\frac{2\sqrt{3}}{3}$
(B) $\frac{2}{3}$
(C) $\frac{\sqrt{3}}{2}$
(D) $\frac{2\sqrt{2}}{3}$
(E) $\frac{3}{2\sqrt{3}}$

Open Ended Questions (FRQ)

Q9. Rationalize the denominator of $\frac{2x}{\sqrt{3x}}$ and simplify. Show all work and explain each step.

Q10. A space probe is traveling at $\frac{3}{\sqrt{2}}$ times the speed of a satellite. Rationalize the denominator and simplify to find the speed of the space probe in terms of the satellite's speed.

Chef's Tips

- Emphasize the importance of rationalizing denominators in scientific calculations.
- Remind students to simplify their answers after rationalizing the denominator.
- Encourage students to use real-world examples, such as space exploration, to illustrate the concept of rationalizing denominators.